

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Hostel Maintenance & Issue Ticketing System

Problem Statement #07 | PES University – Dept. of CSE

1. Jira Backlog – Epics and User Stories

The backlog was set up in Jira using a Company-managed Scrum project. The Lab 1 functional requirements for the Hostel Maintenance & Issue Ticketing System were broken down into 5 Epics and 14 User Stories. Each User Story was linked to its respective Epic.

The following Epics were created:

- EPIC-1: Ticket Submission & Reporting (High)
- EPIC-2: Warden Dashboard & Dispatch (High)
- EPIC-3: SLA Monitoring & Escalation (High)
- EPIC-4: Resident Closure & Feedback (Medium)
- EPIC-5: Reporting & Analytics (Low)

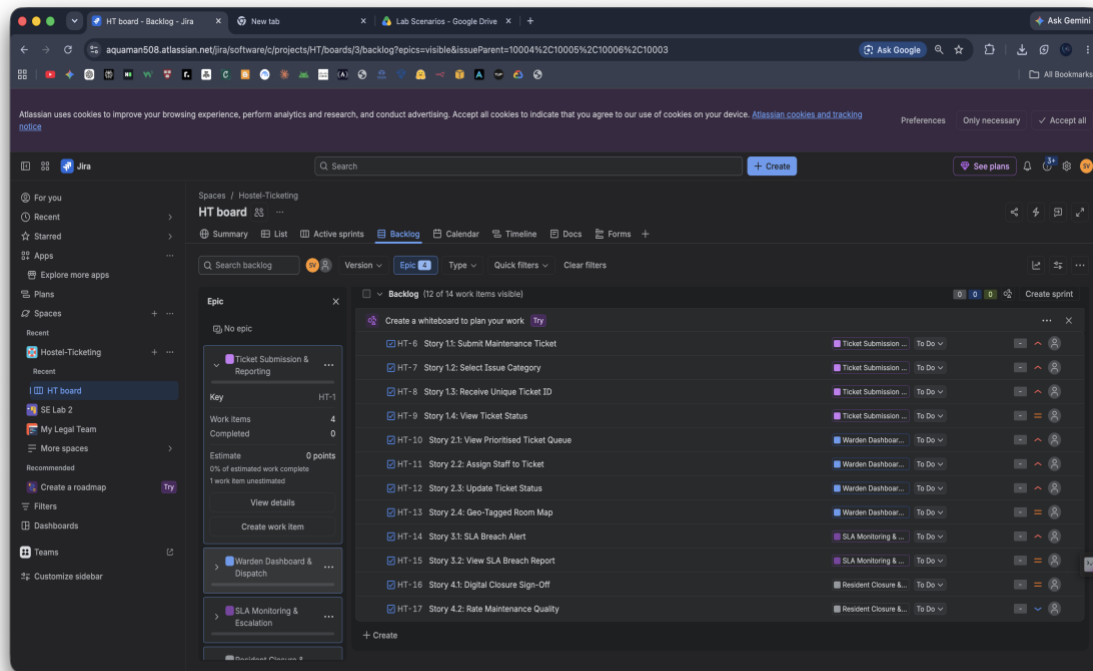


Figure 1 – Jira Backlog showing all Epics and User Stories

2. Story Point Assignments

Story points were assigned to each User Story using the Fibonacci sequence (1, 2, 3, 5, 8, 13...) to estimate the effort, complexity, and uncertainty involved. The team used Planning Poker to agree on estimates.

Story ID	Story Title	Story Points	Priority
HT-6	Story 1.1: Submit Maintenance Ticket	5	High
HT-7	Story 1.2: Select Issue Category	3	High
HT-8	Story 1.3: Receive Unique Ticket ID	2	High
HT-9	Story 1.4: View Ticket Status	3	Medium
HT-10	Story 2.1: View Prioritised Ticket Queue	5	High
HT-11	Story 2.2: Assign Staff to Ticket	3	High
HT-12	Story 2.3: Update Ticket Status	3	High
HT-13	Story 2.4: Geo-Tagged Room Map	8	Medium
HT-14	Story 3.1: SLA Breach Alert	5	High
HT-15	Story 3.2: View SLA Breach Report	5	Medium
HT-16	Story 4.1: Digital Closure Sign-Off	3	Medium
HT-17	Story 4.2: Rate Maintenance Quality	2	Low
HT-18	Story 5.1: Monthly Ticket Report	5	Low
HT-19	Story 5.2: Export Ticket Data as CSV	3	Low

3. Sprint Board – Active Sprint View

Sprint 1 was created with a 1-week duration. The 7 highest priority User Stories were selected and added to Sprint 1. The sprint was started and all stories were moved from To Do to In Progress and then to Done to simulate task completion.

Sprint 1 Goal: Complete core ticket submission and warden dispatch pipeline.

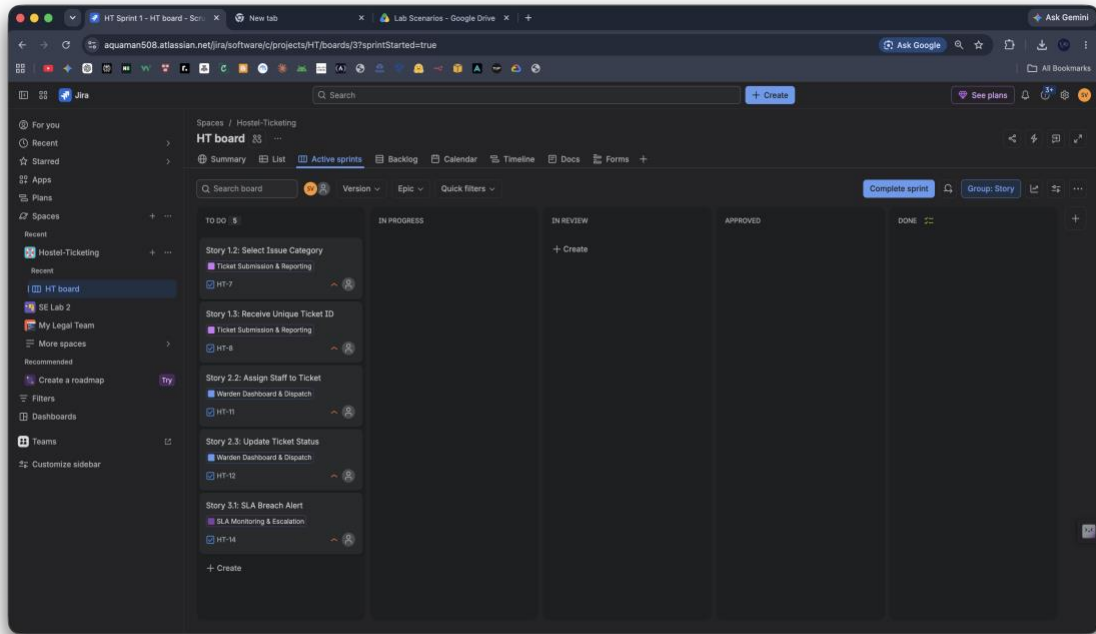


Figure 2 – Active Sprint board (HT Sprint 1) showing stories moved to Done

After completing Sprint 1, Sprint 2 was created with the remaining 5 medium and low priority stories. Sprint 2 Goal: Complete resident feedback and SLA reporting features.

4. Burndown Chart

The Burndown Chart was accessed from Reports > Burndown Chart. The chart tracks the total work remaining in the sprint over time. The grey guideline shows ideal linear progress and the red line shows actual remaining work.

Sprint 1 – Burndown Chart

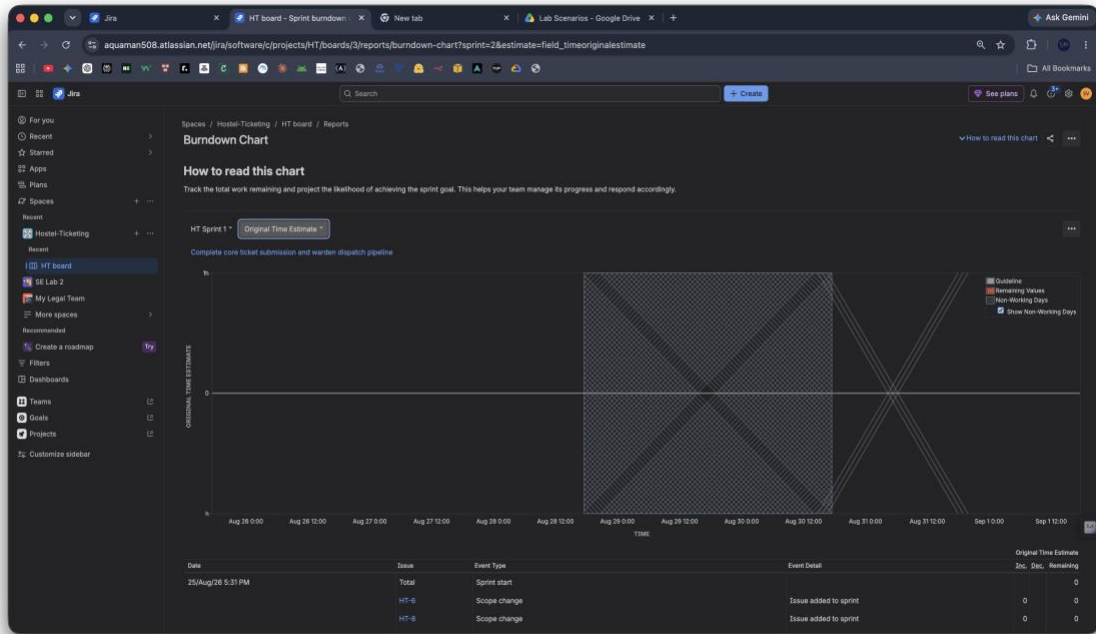


Figure 3 – Burndown Chart for HT Sprint 1

Sprint 2 – Burndown Chart

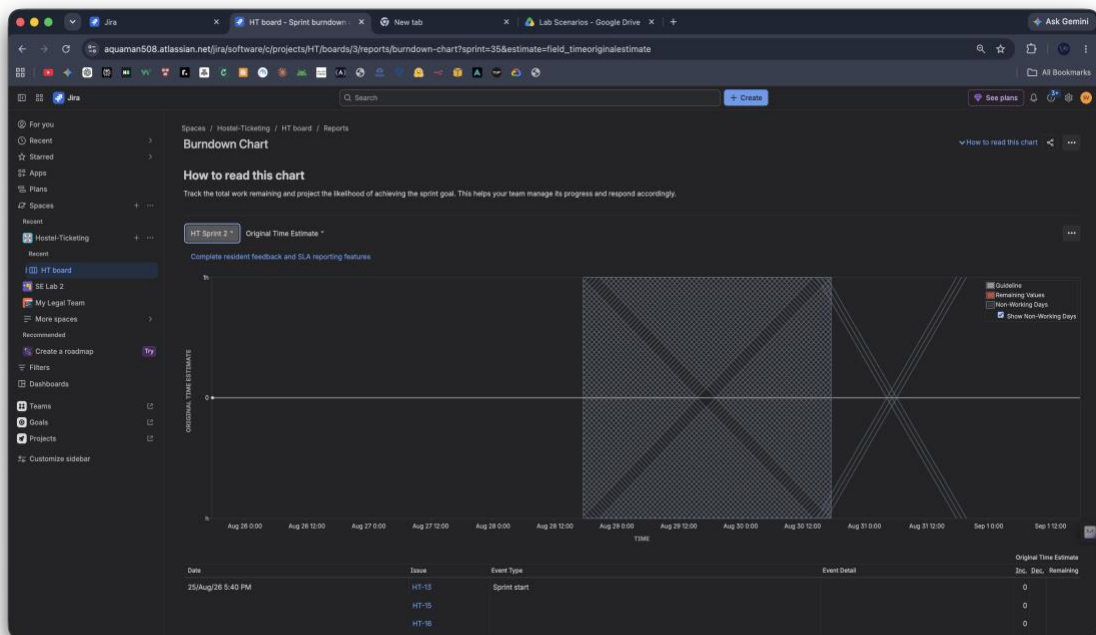


Figure 4 – Burndown Chart for HT Sprint 2

5. Reflection Questions

Q1. Did your estimations reflect the actual effort?

For most stories, the estimates were reasonable. Simple stories like receiving a ticket ID (2 points) were completed quickly as expected. The geo-tagged map story (8 points) took more effort, which the higher point value correctly indicated. Overall the Fibonacci scale helped us think about complexity in relative terms rather than exact hours.

Q2. Was your backlog well-prioritized?

Yes. We put all the High priority stories into Sprint 1 since they covered the most important features – ticket submission, warden dispatch, and SLA alerts. The Medium and Low priority stories like ratings and CSV export were kept for Sprint 2 and later. This made sure the core system worked before adding extra features.

Q3. How did your simulated sprint align with your plan?

Sprint 1 went as planned. All 7 stories were moved from To Do to In Progress to Done without any issues. Sprint 2 also completed all 5 stories. Since this was a simulation done in one session, it was faster than a real sprint, but the workflow and process was clear and easy to follow using the Jira Scrum board.

Q4. What insights did the burndown chart give about your team's capacity?

The burndown charts showed that all stories were completed within the sprint window. In a real sprint the line would go down gradually each day, but since we simulated it in one go it dropped sharply at the end. From this we estimated our team velocity to be around 26 story points for Sprint 1 and 21 for Sprint 2. This gives us a good baseline for planning future sprints.